

UNIT 1: COMPUTER SOFTWARE AND HARDWARE

CHAPTER 1: COMPONENTS OF COMPUTER SYSTEM

INTENDED LEARNING OUTCOMES (ILOS)

AT THE END OF THIS LESSON, THE STUDENT IS EXPECTED TO:

- ❑ Know what computer system is;
- ❑ Differentiate the components of computer system;
- ❑ Enumerate the different types of RAM;
- ❑ Be familiar with common types of storage drivers;
- ❑ List all the types of optical media ;
- ❑ Have a knowledge of different ports and cables

Computer System – A Computer system consists of hard ware and software components.

Hardware is the physical equipment.

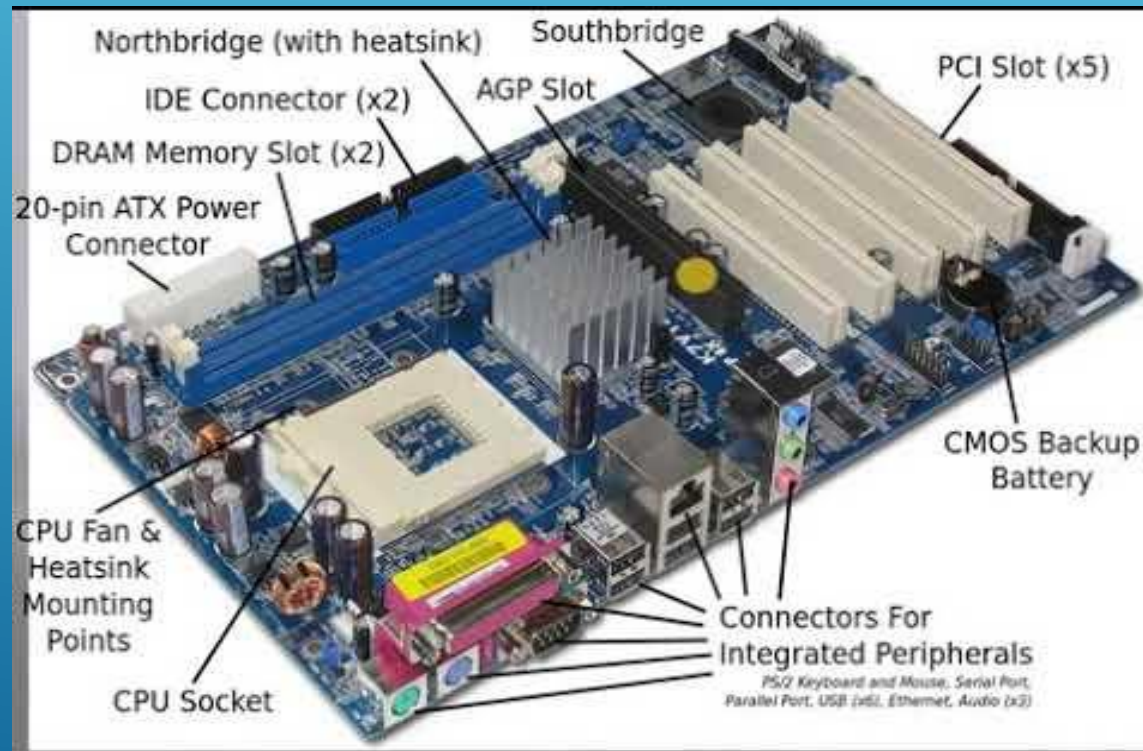
Operating System (OS) instructs the computer how to operate.

Application software performs different functions or program with specific task.

Programs vary widely depending on the type of information that will be accessed or generated.

COMPUTER HARDWARE

a. Motherboard – The motherboard is the main printed circuit board and contains the buses, or electrical pathways, found in a computer. These buses allow data to travel between the various components that comprise a computer. A motherboard is also known as the system, the backplane, or the main board.



b. Central Processing Unit (CPU) – The CPU executes a program, which is a sequence of stored instructions. Each model of processor has an instruction set, which it executes. The CPU executes the program by processing each piece of data as directed by the program and the instruction set.



- ❖ Reduced Instruction Set Computer (RISC) – Architectures use a relatively small set of instructions, and RISC chips are designed to execute these instructions very rapidly.
- ❖ Complex Instruction Set Computer (CISC) – Architectures use a broad set of instructions, resulting in fewer steps per operation.
- ❖ Overclocking is a technique used to make a processor work at a faster speed than its original specification. Overclocking is not a reliable way to improve computer performance and can result in damaging the CPU.
- ❖ MMX is a set of multimedia instructions built into intel processors MMX enabled microprocessors can handle many common multimedia operations that are normally handled by a separate sound or video card. However, only software especially written to call MMX instructions can take advantage of the MMX instructions set.

- ❖ Single Core CPU – One core inside a single CPU chip that handles all of the processing capability. A motherboard manufacturer may provide to build a powerful, multi-processor computer.
- ❖ Dual Core CPU – Two cores inside a single CPU chip in which both cores can process information at the same time.

c. Random-Access Memory (RAM) – It is the temporary storage for data and programs that are being accessed by the CPU.



Different Types of RAM:

- ❖ Dynamic Random Access Memory (DRAM) – Is a memory chip that is used as main memory. DRAM must be constantly refreshed with pulses of electricity in order to maintain the data stored within a chip.
- ❖ Static Random Access Memory (SRAM) – Is a memory chip that is used as cache memory. SRAM is much faster than DRAM and does not have to be refreshed as often.
- ❖ Fast Page Mode DRAM (FPM Memory) – Is memory that supports paging. Paging enables faster access to the data than regular DRAM. Most 486 and Pentium systems from 1995 and earlier use FPM memory.
- ❖ Extended Data Output RAM (EDO Memory) – Is memory that overlaps consecutive data accesses. This speeds up the access time to retrieve data from memory, because the CPU does not have to wait from one data access cycle to end before another data access cycle begins.

- ❖ Synchronous DRAM (SDRAM) – Is DRAM that operates in synchronization with the memory bus. The memory bus is the data path between the CPU and the main memory.
- ❖ Double Data Rate 2 SDRAM (DDR2 SDRAM) – Is faster than DDR SDRAM memory. DDR2 SDRAM improves performance over DDR SDRAM by decreasing noise and crosstalk between the signal wires.
- ❖ Rambus DRAM (RDRAM) – Is a memory chip that was developed to communicate at very high rates of speed. RDRAM chips are not commonly used.

d. Power Supply – The power supply unit is the piece of hardware that's used to convert the power provided from the outlet into usable power for the many parts inside the computer case



e. Video Card – A video card is a piece of computer hardware that's numerous contacts on the bottom of the card and one or more ports on the side for connection to video displays and other devices.



f. Hard Drives and Floppy Drives – A storage drive reads or writes information to magnetic or optical storage media. The drive can be used to store data permanently or to retrieve information from a media disk.

Here are Some Common Types of Storage Drives:

- A floppy drive, or floppy disk drive, is a storage device that uses removable 3.5-inch floppy disk.
- A hard drive, or hard disk drive, is a magnetic storage device that is installed inside the computer.



- An optical drive is a storage device that uses lasers to read data on the optical media. There are two types of optical drives CD and DVD media which can be pre-recorded (read-only), recordable (write once), or re-recordable (read and write multiple times).
These are several types of optical media:

- CD-ROM – CD read only memory media that is pre-recorded.
- CD-R – CD-recordable media that can be recorded once.
- CD-RW – CD-rewritable media that can be recorded, erased, and recorded
- DVD-ROM – DVD read-only memory media that is pre-recorded



- DVD-RAM – DVD-random access memory media that can be recorded, erased, and re-recorded.
- DVD+/-R – DVD-recordable media that can be recorded once.
- DVD+/-RW – DVD-rewritable media that can be recorded, erased, and re-recorded.

Flash drive, also known as a thumb drive, is a removable storage device that connects to a USB port. A flash drive uses a special type of memory that requires no power to maintain the data.



Hard drives and optical drives are manufactured with different interfaces that are used to connect the drive to the computer. To install a storage drive in a computer, the connection interface on the drive must be the same as the controller on the motherboard.

Here are some common drive interfaces:

- Integrated Drive Electronics (IDE) – also called Advanced Technology Attachment (ATA) is an early drive controller interface that connects computers and hard disk drives. An IDE interface uses a 40-pin connector.
- Enhanced Integrated Drive Electronics(EIDE) – also called ATA-2, is an updated version of the IDE drive controller interface. EIDE supports hard drive larger than 512MB, enables Direct Memory Access (DMA) for speed, and uses the AT Attachment Packet interface (ATAPI) to accommodate optical drives and tape drives on the EIDE bus. An EIDE interface uses a 40-pin connector.
- Parallel ATA (PATA) – refers to the parallel version of the ATA drive controller interface.
- Serial ATA (SATA) – refers to the serial version of the ATA drive controller interface. A SATA interface uses a 7-pin connector.

Small Computer System Interface (SCSI) – is a drive controller interface that can connect up to 15 drives. SCSI connect both internal and external drives. An SCSI interface uses a 50-pin, 68-pin, or 80-pin connector.

PORTS AND CABLES

- a. Serial Ports and Cables – A serial port can be either a DB-9 or a DB-25 male connector. Serial ports transmit one bit of data at a time.
- b. USB Ports and Cables – The Universal Serial Bus (USB) is a standard interface that connects peripheral devices to a computer
- c. FireWire Ports and Cables – FireWire is a high-speed, hot-swappable interface that connects peripheral devices to a computer.
- d. Parallel Ports and Cables – A parallel port on a computer is a standard Type A DB-25 female connector, The parallel connector on a printers may use a Type C high-density 36-pin connector.

e. SCSI Ports and Cables – a SCSI port can transmit data at rates in excess of 320Mbps and can support up to 15 devices. If a single SCSI device is connected to an SCSI port, the cable can be up to 80 feet (24.4m) in length.

- DB-25-female connector, High-density-50-pin female connector
- High-density 68-pin female connector

f. Network Ports and Cables – A network port, also known as an RJ-45 port, connects a computer to a network. The connection speed depends on the type of network port.

g. PS/2 Ports – a PS/2 port connects a keyboard or a mouse to a computer. The PS/2 port is a 6-pin mini-DIN female connector.

h. Audio Ports – An audio port connects audio devices to the computer.

- Line In – Connects to an external source, such as a stereo system.
- Microphone – Connects to a microphone
- Line Out – Connects to speakers or headphones
- Game port/MIDI – Connects to a joystick or MIDI-interfaced device

I. Video Ports and Connectors – A video port connects a monitor cable to a computer. There are several video port and connector types:

- Video Graphics Array (VGA) – VGA has a 3-row 15-pin female connector and provides analog output to a monitor.
- High Definition Multimedia Interface (HDMI) – HDMI has a 19-pin connector and provides digital video and digital audio signals.
- Component/RGB – RGB has three shielded cables (red, green, blue) with RCA jacks and provides analog video signals.